JUAN SEBASTIÁN SÁNCHEZ URBANO

Industrial Automation Engineer · Machine Learning, Technical Sales, Power BI and AI

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LinkedIn · GitHub · Portfolio

PROFESSIONAL PROFILE

Industrial Automation Engineer focused on Machine Learning, technical sales and data analysis. Since October 2026 I have been a technical advisor at Industrias CATO, where I advise customers on fastener selection and technical solutions. Previously I worked as a technical and sales advisor at a technical-solutions company, where I analyzed customer information and automated data processes, and as CNC and Robotics area leader I analyzed production and quality-control variables. I build predictive models, Power BI dashboards and AI workflows with Python, R and n8n, and design medallion-architecture (Bronze/Silver/Gold) ETL pipelines. I combine industrial engineering with analytics to turn data into sales and production decisions. I maintain a public portfolio with projects, my own tools and automation, data and 3D printing services.

TECHNICAL SKILLS

Machine Learning & AI: Machine Learning, Predictive modeling, Classification, Statistical analysis, EDA, AI agents, AI chatbots, Python, R.

Sales & Business Intelligence: Technical and commercial advisory, Sales analysis, Power BI, Advanced Excel, Dashboards, Data consolidation.

Data engineering & Cloud: ETL pipelines, Medallion architecture, Advanced SQL, MySQL, AWS, REST APIs, Data quality and cleaning.

Automation & Industrial engineering: n8n, Industrial automation, PLC programming, CNC and robotics, Real-time IoT sensors.

Web development & deployment: HTML, CSS, JavaScript, Bash, YAML, Git and GitHub, GitHub Actions, GitHub Pages.

Methodology & Management: PDCA (continuous improvement), Analytical problem solving, Results orientation, Effective communication.

PROFESSIONAL EXPERIENCE

Technical Advisor — Industrias CATO

Oct 2026 – Present

- I advise customers on fastener selection: threads, heads, materials, finishes and applications, and support the technical validation of their requirements.
- **Achievement:** I built technical reference tools available in this portfolio: a fastener academy and a thread calculator.

Technical, Sales and Data Automation Advisor — Technical solutions company

2019 – Nov 2025 · Mar 2026 – Sep 2026

- I analyze and validate customers' technical and operational information to structure comprehensive solutions, ensuring the quality and traceability of commercial records.
- **Achievement:** I designed and implemented an AI chatbot for the web channel, integrated into an n8n automation pipeline that automatically extracts, validates and routes reports and information to the responsible areas.

CNC and Robotics Area Lead (Production Data Analysis) — Metálicas Mundial

Oct 2025 – Jan 2026

- I led the capture and analysis of critical production and quality-control variables, ensuring the reliability of manufacturing process data.
- **Achievement:** I identified bottlenecks and optimized cycle times by analyzing machinery operating data under agile continuous-improvement methodologies, increasing the area's productivity.

FEATURED PROJECTS

End-to-end automated ETL pipeline

MinTIC / Talento Tech · 2026 · Python, n8n, REST APIs, ETL

- An extraction, transformation and loading architecture with Python, n8n and REST API calls to consolidate centralized databases.
- **Impact:** 100% of manual data entry eliminated, with complete, standardized and reliable data for executive analysis.

SIVIGILA epidemiological ETL 2018–2022

Dengue · SARI · Tuberculosis · Python, Excel, Medallion, HTML dashboard

- An ETL process over five National Institute of Health Excel files, with extraction, transformation and loading scripts, medallion architecture and an HTML dashboard to visualize results.

Kaismart Solutions ETL lab

Universidad Autónoma de Occidente · MySQL, Python, Excel, Medallion, Schedule

- Integration of a MySQL sales table (5,000 records) with 50,000 logistics events in Excel, joined by order ID. Includes exploratory analysis, medallion architecture, 10 business questions and scheduled execution.

Web portfolio with medallion architecture

2026 · HTML, CSS, JavaScript, Python, GitHub Actions

- Site published on GitHub Pages with a repository in Bronze/Silver/Gold layers, an automated Python build and continuous deployment with GitHub Actions. Includes my own tools: an HTML reader with safe preview, a WhatsApp link generator and a JSON formatter.

Real-time data capture and alerts (industrial IoT)

Engineering projects · 2022 – 2024 · IoT, Sensors, Alerts, Real time

- Integration of industrial sensors and instrumentation for real-time data ingestion, automatic validation of critical operating variables and timely control alerts.

3D Workshop and automation services

Business venture · 3D printing, n8n, AI, WhatsApp

- Custom products (printed keychains and cases, 3D printing) and automation services with n8n, AI content and industrial engineering, with a product designer connected to WhatsApp.

EDUCATION

Industrial Automation Engineering

Graduate engineer